Limitations and responsible use

What the target does—and does not—mean

`has_risk` is a supplied historical proxy label. The reported metrics measure a model's agreement with that label, not verified ingredient presence, toxicity, consumer safety, medical risk, personal allergy, legal compliance, or product suitability. A negative prediction is not a certificate that a formula is safe or free from every relevant substance.

The source metadata provides a **count of 16** reference substances only. The exact roster, names, aliases, source versions, original matching rules, and source labeling code are unavailable. No undocumented list is reconstructed. Consequently, no deterministic matching baseline is evaluated and no masking is applied.

Data and scope limitations

- The historical source has noisy/missing metadata, multilingual names, possible OCR/transcription artifacts, and imperfect product identity information.
- A French-language export is not proof of a France-only source population; country tags can be multi-valued.
- The primary cohort is not a curated legal or regulatory cosmetics-only population. The 6,785-row recognized category subset is operational, may include oral-care categories, and is not a legal classifier.
- Cleaning quarantines 47 low raw/list-alignment rows. This data-quality heuristic can still miss or misidentify real source issues; long/prose-like lists are flagged rather than resolved by inference.
- The model lacks concentration, formulation order meaning, exposure route/dose/duration, individual sensitivity, manufacturing context, and evolving regulatory status.

Validation limitations

Exact normalized formula groups do not cross the 12,459/3,115 internal split, but grouping cannot prevent all near-duplicate leakage or establish performance for future/external product populations. Model choice, tuning, and threshold selection rely on training-side grouped CV. The primary test is an internal holdout, not external validation.

If present, the five-fold recognized-scope result overlaps the source population and selected-model information. It is a sensitivity analysis, not independent external validation, a confidence interval, or fully nested model selection.

Model and interpretation limitations

The Random Forest learns binary ingredient-presence patterns plus `log1p(ingredient_count)`. It omits categories and brands by design but can still inherit correlations and label artifacts from supplied ingredient patterns. Performance can differ by omitted subgroups and data quality conditions.

MDI global importance is non-negative split-usage importance. It contains no direction of association and is not an ingredient toxicity ranking, causal conclusion, or safety endorsement. Local drop-one probability perturbations are model-behavior checks; they are **not SHAP** and cannot establish scientific causality.

Deployment and governance limits

The application is local only. No hosted service, OCR ingestion, production monitoring, authentication, clinical workflow, full regulatory workflow, or incident-response process is claimed. Model files should be treated as trusted-code artifacts only.

Responsible next steps

Before any stronger claim, obtain and version the actual registry and label-generation rules, audit matching behavior, acquire independent labels/populations, evaluate calibration and subgroup behavior, and conduct an appropriate domain/regulatory review. Review, understand, reproduce, and disclose assisted code in accordance with cohort policy before submitting or sharing the work.